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# **Improving Sim-to-Real transfer of Learning-based Policies for Quadcopter Racing through Domain Randomization**

**Semestral Project**

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## **Abstract**

The study of autonomous Unmanned Aerial Vehicles (UAVs) has become a prominent sub-field of mobile robotics.

**Keywords** Unmanned Aerial Vehicles, Automatic Control

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## Abbreviations

**UAV** Unmanned Aerial Vehicle

**RL** Reinforcement Learning

**DR** Domain Randomization

**POMDP** Partially Observable Markov Decision Process

**DoF** Degrees of Freedom

**Sim2Real** Simulation-to-Reality

**PPO** Proximal Policy Optimization

**AC** Actor Critic

**MLP** Multi-Layer Perceptron

**CTBR** Collective Thrust and Body Rates

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## ■ 1 Introduction

In the context of drone racing, where quadcopters are pushed to their physical and computational limits, there is a growing trend toward using neural networks for control. As demonstrated in [1], this approach has made possible to outperform some of the world's most skilled human pilots.

These neural networks controllers are typically optimized using RL algorithms. However, despite many of the advantages that RL bring, it is still a highly sample inefficient algorithm that makes collecting data in the real world dangerous and expensive. Because of this, training is often performed in simulated environments that provide a safe and scalable alternative. Nonetheless, this alternative leads to other problems since the resulting policies, learned in simulations, often perform worse than expected or fail to transfer to real-world scenarios altogether. This discrepancy in the policy's performance between simulation and reality is known as the Simulation-to-Reality (Sim2Real) gap, and it mainly arises from an undesired overfit of the optimization process to the simulated scenario.

### ■ 1.1 Related works

Demonstrating that Domain Randomization (DR) can significantly increase the adaptability and robustness of autonomous systems, the work of [2] serves as the direct inspiration for this project. Moreover in [3], the authors show how an appropriate system identification combined with DR effectively bridges the Sim2Real gap.

## ■ 2 Methodology

### ■ 2.1 Problem definition

We define a domain as a specific subset of possible Partially Observable Markov Decision Process (POMDP) environments. While a complete POMDP is formally characterized by an 8-tuple [4], we focus on a reduced 6-tuple representation that captures the essential elements for our domain definition. A standard domain is then characterized by  $\mathcal{D} = (\mathcal{S}, \mathcal{A}, \mathcal{O}, P, O, R)$  where:

- $\mathcal{S}$  : denotes a set of states called the state space.
- $\mathcal{A}$  : is the set of actions called the action space.
- $\mathcal{O}$  : is the set of observations called the observation space.
- $\mathbf{P}(\mathbf{s}'|\mathbf{s}, \mathbf{a})$  : is the probability that a given action  $a \in \mathcal{A}$ , at state  $s \in \mathcal{S}$  will lead to a new state  $s' \in \mathcal{S}$ .
- $\mathbf{O}(\mathbf{o}|\mathbf{s}', \mathbf{a})$  : is the observation probability, representing the probability of observing  $o \in \mathcal{O}$  after taking action  $a$  and transitioning to state  $s'$ .
- $\mathbf{R}(\mathbf{s}', \mathbf{s}, \mathbf{a})$  : is the immediate reward, or expected immediate reward, received for transitioning from  $s$  to  $s'$  after performing action  $a$ .

From the source domain,  $\mathcal{D}_s$ , (i.e. simulation), we can control a set of  $N$  parameters that define a configuration vector,  $\boldsymbol{\xi} = [\xi_i, \xi_{i+1}, \dots, \xi_N]$ . These parameters will affect the transition probability  $\mathbf{P}(\mathbf{s}'|\mathbf{s}, \mathbf{a})$  of the domain. Examples of  $\xi_i$  can include:

|                       |                                                                                    |
|-----------------------|------------------------------------------------------------------------------------|
| Physical properties   | $\xi_{\text{mass}}, \xi_{\text{inertia}}, \xi_{\text{drag}}, \xi_{\text{gravity}}$ |
| Perception properties | $\xi_{\text{sensor\_noise}}, \xi_{\text{sensor\_latency}}$                         |
| Visual properties     | $\xi_{\text{lighting}}, \xi_{\text{texture}}, \xi_{\text{fov}}$                    |
| Scene properties      | $\xi_{\text{gate\_pos}}, \xi_{\text{gate\_size}}, \xi_{\text{gate\_num}}$          |

When DR is applied, each configuration  $\boldsymbol{\xi}$  is sampled from a bounded randomization space  $\Xi$  with a probability distribution. That is to say,  $\boldsymbol{\xi} \in \Xi \subset \mathbb{R}^N$ ,  $\boldsymbol{\xi} \sim p(\boldsymbol{\xi})$  where  $\xi_i \in [\xi_i^{\text{low}}, \xi_i^{\text{high}}]$ . At the start of every training episode, a new environment instance is created by sampling a new configuration vector  $\boldsymbol{\xi}$  from the distribution  $p(\boldsymbol{\xi})$ . Therefore, the agent will interact with a new POMDP environment, a new domain  $\mathcal{D}_{\boldsymbol{\xi}}$ . This way, the agent can collect data from a variety of environments that otherwise would remain unexplored.

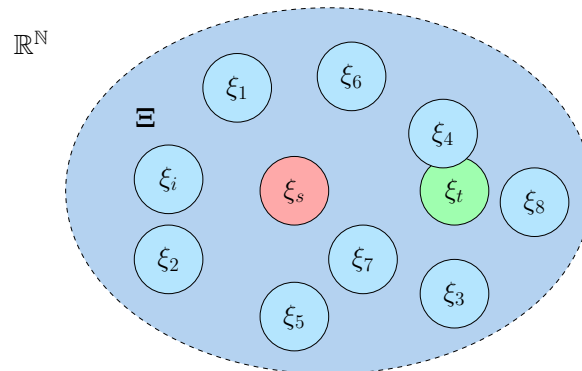


Figure 2.1: In red, the source domain randomization configuration. In light blue, randomized instance of the source configuration. In green, the target domain configuration.

Hereby, during training, the neural network weights and biases  $\theta$  are optimized to maximize the expected reward  $R$  **across a distribution of randomized domains** [5]. With the aim of finding the optimal,  $\theta^*$  that will lead to the optimal parametrized policy  $\pi_\theta^*$ .

$$\theta^* = \arg \max_{\theta} \mathbb{E}_{\xi \sim \Xi} [\mathbb{E}_{\pi_\theta, \tau \sim D_\xi} [R(\tau)]] \quad (2.1)$$

Where  $R(\tau) = \sum_{t=0}^n \gamma^t r_t$  is the cummulative discounted reward over a trajectory (or sequence) of states and actions  $\tau = (s_0, a_0, s_1, a_1, \dots)$ , for the episode length  $n$ .

As illustrated in Figure 2.1 and described in Equation 2.1, the policy is exposed to a variety of domains to promote generalization and improve robustness, increasing the likelihood of successful transfer to the real environment. However, this strategy further reduces the sample efficiency of RL and increases training time. Therefore, it is essential to properly bound the randomization space  $\Xi$  so that it encompasses only the minimum necessary range to include the real domain.

## ■ 2.2 Quadcopter model

In the simulated environments, the quadcopter is modeeld as a 6 Degrees of Freedom (DoF), rigid body. The state of the quadcopter is defined by its position  $\mathbf{p}$ , velocity  $\mathbf{v}$ , orientation  $\mathbf{q}$ , body rates  $\boldsymbol{\omega}$  and rotor speeds  $\boldsymbol{\Omega}$ . Namely,  $\mathbf{x} = [\mathbf{p}, \mathbf{v}, \mathbf{q}, \boldsymbol{\omega}, \boldsymbol{\Omega}]^T$ . The position, velocity and acceleration are respresented in the world frame  $\mathcal{W} \in \mathbb{R}^3$ , while the body rates in the body frame  $\mathcal{B} \in \mathbb{R}^3$ . The orientation is described by an unit quaternion rotation  $\mathbf{q} \in \mathbb{SO}(3)$ . Hence, the kinematics and dynamics of the drone are:

$$\dot{\mathbf{p}}^{\mathcal{W}} = \mathbf{v}^{\mathcal{W}} \quad (2.2)$$

$$\dot{\mathbf{q}} = \frac{1}{2} \mathbf{q} \odot [0 \quad \boldsymbol{\omega}^{\mathcal{B}}]^T \quad (2.3)$$

$$\dot{\mathbf{v}}^{\mathcal{W}} = \frac{R(\mathbf{q})}{m} (\mathbf{f}_T^{\mathcal{B}} + \mathbf{f}_D^{\mathcal{B}}) + \mathbf{g}^{\mathcal{W}} \quad (2.4)$$

$$\dot{\boldsymbol{\omega}}^{\mathcal{B}} = J^{-1} (\boldsymbol{\tau}^{\mathcal{B}} - \boldsymbol{\omega}^{\mathcal{B}} \times J \boldsymbol{\omega}^{\mathcal{B}}) \quad (2.5)$$

$$\dot{\boldsymbol{\Omega}} = \frac{1}{k_m} (\boldsymbol{\Omega}_c - \boldsymbol{\Omega}) \quad (2.6)$$

Where,  $\odot$  represents quaternion multiplication,  $R(\mathbf{q})$  is the rotational matrix from body frame  $\mathcal{B}$  to world frame  $\mathcal{W}$  at orientation  $\mathbf{q}$ ,  $J$  is the diagonal inertia matrix,  $\mathbf{f}_T^{\mathcal{B}}$  and  $\mathbf{f}_D^{\mathcal{B}}$  are the collective thrust and drag respectively,  $k_m$  the motor time constant,  $m$  the qudcopter's mass and  $\mathbf{g}$  is gravity.

In the body frame  $\mathcal{B}$ , the individual motor thrusts  $T_i$ ,  $i = 1, \dots, 4$ , are assumed to be perpendicular to the  $\mathbf{xy}$  plane. The collective thrust  $\mathbf{f}_T^{\mathcal{B}}$ , and torques  $\boldsymbol{\tau}^{\mathcal{B}}$ , can be maped from the individual motor thrusts with the allocation matrix  $\Gamma$ .

$$\begin{bmatrix} \mathbf{f}_T^{\mathcal{B}} \\ \tau_x \\ \tau_y \\ \tau_z \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ d/\sqrt{2} & -d/\sqrt{2} & -d/\sqrt{2} & d/\sqrt{2} \\ -d/\sqrt{2} & -d/\sqrt{2} & d/\sqrt{2} & d/\sqrt{2} \\ c_{tf} & -c_{tf} & c_{tf} & -c_{tf} \end{bmatrix} \begin{bmatrix} T_1 \\ T_2 \\ T_3 \\ T_4 \end{bmatrix} \quad (2.7)$$

Being  $d$  the arm length and  $c_{tf}$  the linear torque constant. Likewise, the indidual motror thrusts  $T_i$ , can be mapped from the rotor speeds  $\Omega_i$  with the thrust map coefficients,  $\mathbf{k}_T = [a_0, a_1, a_2]$  that are determined experimentally fitting a quadratic curve.

$$T_i = \mathbf{k}_T \begin{bmatrix} \Omega_i^2 \\ \Omega_i \\ 1 \end{bmatrix} = [a_0 \quad a_1 \quad a_2] \begin{bmatrix} \Omega_i^2 \\ \Omega_i \\ 1 \end{bmatrix} \quad (2.8)$$

The total drag force,  $\mathbf{f}_D^{\mathcal{B}}$  is calculated using a simplified aerodynamic model function of the dron's linear velocity and proportional to the drag coefficients.

$$\mathbf{f}_D^{\mathcal{B}} = \begin{bmatrix} C_{D_x} & 0 & 0 \\ 0 & C_{D_y} & 0 \\ 0 & 0 & C_{D_z} \end{bmatrix} \begin{bmatrix} v_x^{\mathcal{B}} \\ v_y^{\mathcal{B}} \\ v_z^{\mathcal{B}} \end{bmatrix} \quad (2.9)$$

## ■ 2.3 RL Set Up

Although the simplified quadcopter model defined in 2.2 is used to represent the drone dynamics in simulation, the RL setup employed in this project is **model-free**. Specifically, it consists of an Actor Critic (AC) structure optimized using Proximal Policy Optimization (PPO) [6]. The implementation relies on Stable-Baselines3 [7] to interface with a custom C++ Gymnasium environment (based on flightmare [8]). To train the AC networks PPO is used.

The actor policy network  $\pi_\theta$ , is defined by a Multi-Layer Perceptron (MLP) that takes an observation vector at time step  $t$ , of 31 inputs (see Equation 2.10). The observation vector is sampled from a bounded observation space  $o_t \in \mathcal{O} \subset \mathbb{R}^{31}$ .

$$o_t = [\mathbf{p}_t \quad \mathbf{v}_t \quad \text{vec}R(\mathbf{q})_t \quad u_{t-1} \quad \mathbf{p}_{\mathbf{g}_t}^{TL} \quad \mathbf{p}_{\mathbf{g}_t}^{TR} \quad \mathbf{p}_{\mathbf{g}_t}^{BL} \quad \mathbf{p}_{\mathbf{g}_t}^{BR} \quad \mathbf{p}_t^{\text{look}}]^T \quad (2.10)$$

Where  $u_{t-1}$  is the throttle motor command at the previous time step.  $\mathbf{p}_{\mathbf{g}_t}^X$  are relative positions of the corners of the target gate at time  $t$  with respect the body frame  $\mathcal{B}$ . And,  $\mathbf{p}_t^{\text{look}}$  is the look at position of the target gate at time  $t$ , also in the body frame. The input layer is followed by three hidden layers of 128 neurons each, and it outputs 4 neurons. The outputs are the Collective Thrust and Body Rates (CTBR), sampled from a bounded action space  $a_t \in \mathcal{A} \subset \mathbb{R}^4$ .

$$a_t = [\mathbf{f}_{Tt} \quad \boldsymbol{\omega}_t]^T \quad (2.11)$$

On the other hand, the critic network architecture  $V_f$ , is formed also by 3 hidden layers but of 256 neurons each, it has the same input space and it outputs a single value based on the observation  $o_t$ .

## ■ Reward System

## ■ 2.4 DR configuration

The randomization configuration considered for this semestral project has been the following:

$$\boldsymbol{\xi}_d = [m, J, C_D, k_m, \mathbf{k}_T, \Gamma]$$

Every parameter is relatively randomized (multiplicative randomization) by a constant sampled from an uniform distribution  $\mathcal{U}(1 - p, 1 + p)$  parametrized by the parameter  $p$  which indicates the randomization percentage.



### 3 Results

To investigate how DR affects the transferability of controllers learned in simulation to different environments, the following experiments were designed. First, three randomized drone instances were initialized in simulation, and a common policy was executed concurrently in Gazebo during 5 minutes to measure the uav's lap times and success rates. The policies were trained to race in a simple eight-figure track (see Figure 3.1) and their dynamical parameters can be found in Table A.1. This procedure was performed for four distinct policies trained at randomization levels of 0%, 10%, 20%, and 30%. The results of this first experiment are presented in section 3.1.

Following successful validation in the simulated environment, the experiments were replicated on physical hardware with.....

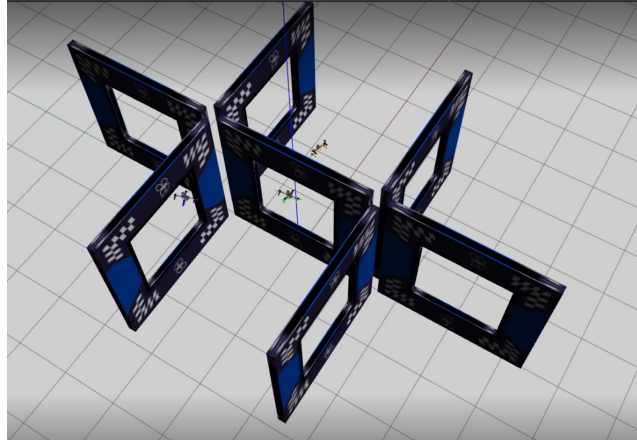


Figure 3.1: Three randomized drones racing in Gazebo.

#### 3.1 Simulation to Simulation Results

As depicted in Figure 3.2, the baseline configuration with 0% randomization yields to the fastest average lap time of 4.13s. However, this speed comes with a significant cost in reliability. More than half of the laps resulted in a crash with a gate and two of the drones

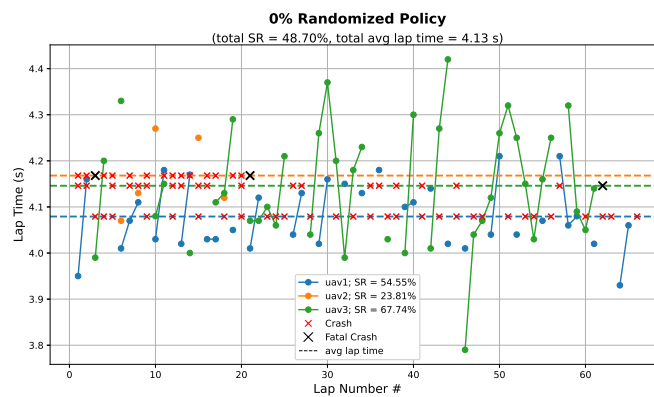


Figure 3.2: 10DR simulation results

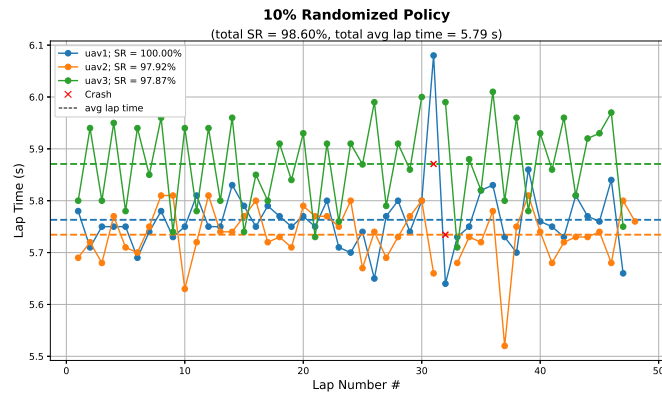


Figure 3.3: 10DR simulation results

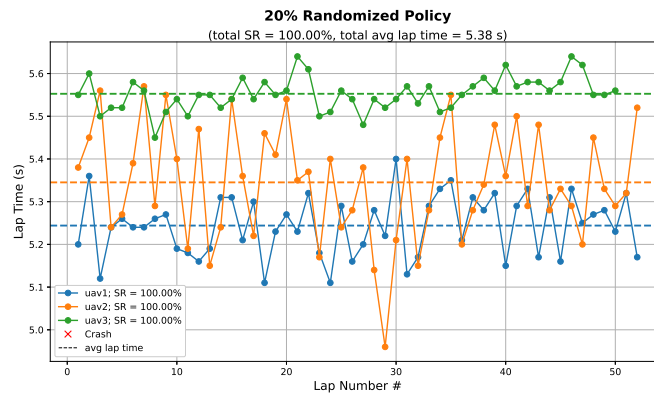


Figure 3.4: 10DR simulation results

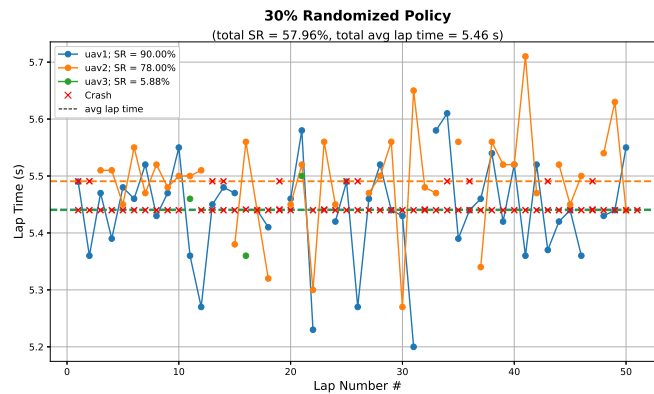


Figure 3.5: 10DR simulation results

### ■ 3.2 Simulation to Reality Results

| Policy | UAV  | Success Rate[%] |      | Number of Laps# |      | Avg Lap Time[s] |      | Fatal Crash |      |
|--------|------|-----------------|------|-----------------|------|-----------------|------|-------------|------|
|        |      | Sim             | Real | Sim             | Real | Sim             | Real | Sim         | Real |
| 0% DR  | a300 |                 |      |                 |      |                 |      |             |      |
|        | uav1 | 54.55           |      | 36              |      | 4.08            |      | No          |      |
|        | uav2 | 23.81           |      | 5               |      | 4.17            |      | <b>Yes</b>  |      |
|        | uav3 | 67.74           |      | 42              |      | 4.15            |      | <b>Yes</b>  |      |
| 10% DR | a300 |                 |      |                 |      |                 |      |             |      |
|        | uav1 | 100             |      | 47              |      | 5.76            |      | No          |      |
|        | uav2 | 97.92           |      | 47              |      | 5.73            |      | No          |      |
|        | uav3 | 100             |      | 46              |      | 5.87            |      | No          |      |
| 20% DR | a300 |                 |      |                 |      |                 |      |             |      |
|        | uav1 | 100             |      | 52              |      | 5.24            |      | No          |      |
|        | uav2 | 100             |      | 52              |      | 5.35            |      | No          |      |
|        | uav3 | 100             |      | 50              |      | 5.55            |      | No          |      |
| 30% DR | a300 |                 |      |                 |      |                 |      |             |      |
|        | uav1 | 90              |      | 45              |      | 5.44            |      | No          |      |
|        | uav2 | 78              |      | 39              |      | 5.49            |      | No          |      |
|        | uav3 | 5.88            |      | 3               |      | 5.44            |      | No          |      |

Table 3.1: (sim time 5 min)

## ■ 4 Conclusion

Summarize the achieved results. Can be similar as an abstract or an introduction, however, it should be written in past tense.

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## ■ A Appendix A

| Parameter                   | Nominal     | Randomized<br>Parameter * $\mathcal{U}(1 - p, 1 + p)$ |               |               |
|-----------------------------|-------------|-------------------------------------------------------|---------------|---------------|
|                             |             | uav1<br>p = 10%                                       | uav2<br>p=20% | uav3<br>p=30% |
| mass [kg]                   | 1.178       | 1.041                                                 | 0.814         | 1.212         |
| body_radius [m]             | 0.05        | 0.964                                                 | 0.883         | 0.702         |
| body_height [m]             | -0.1        | 1.023                                                 | 1.177         | 1.271         |
| mass_prop [kg]              | 0.008       | 1.095                                                 | 1.143         | 1.191         |
| radius_prop [m]             | 0.089408    | 1.072                                                 | 0.918         | 1.288         |
| motor_mesh_z_offset [m]     | -0.0086     | 0.924                                                 | 0.824         | 0.924         |
| rotor_xy_offset [m]         | 0.1         | 1.067                                                 | 1.192         | 1.067         |
| rotor_z_offset [m]          | -0.055      | 0.912                                                 | 1.118         | 0.912         |
| rotor_velocity_slowdown_sim | 0.0159236   | 1.034                                                 | 1.132         | 0.713         |
| motor_constant              | 20.21       | 1.081                                                 | 0.804         | 1.229         |
| moment_constant             | 0.012       | 0.922                                                 | 1.162         | 0.731         |
| time_constant_up [s]        | 1/(20.5761) | 0.953                                                 | 0.886         | 1.221         |
| time_constant_down [s]      | 1/(18.3486) | 1.067                                                 | 1.189         | 0.708         |
| max_rot_velocity            | 1.0         | 1.058                                                 | 0.833         | 1.197         |
| rotor_drag_coefficient      | 0.1         | 1.041                                                 | 1.144         | 1.276         |
| inertia_body_radius [m]     | 0.25        | 1.091                                                 | 0.817         | 1.298         |
| inertia_body_height [m]     | 0.05        | 0.912                                                 | 1.178         | 0.744         |

Table A.1: Parameter multipliers for 10%, 20%, and 30% randomized UAVs. Nominal column retains original values.